Index Number : [redacted] .... Program: ~~Electrical and Electronic Eng.~~

UNIVERSITY OF ENERGY AND NATURAL RESOURCES

SCHOOL OF SCIENCES

DEPARTMENT OF MATHEMATICS AND STATISTICS

Level 100, End of Second Semester Examinations , 2015/2016

B.Sc. (Comp. Eng./Pet. Eng./Ren. En./Agric. Eng./ Elect. Eng./ Mech. Eng./Env. Eng.)

MATH 109: ENGINEERING MATHS 1

18th December, 2015 Time: 2.5 Hours

INSTRUCTION: Answer All Questions in Section A (45 Marks)

1. Fill in the following

   a) $\infty - \infty = \cdots$ .... [0.5 mark]

   b) $e^{\infty} =$ .... [0.5 mark]

   c) $\frac{\infty}{\infty} =$ .... [0.5 mark]

   d) $\frac{0}{0} =$ .... [0.5 mark]

   e) $\frac{1}{\infty} =$ .... [0.5 mark]

2. Which of the following is/ are not true about rational number(s)?

   I. $\sqrt{\frac{112}{7}}$ II. $\sqrt{6}$ III. $0.\dot{3}$

   a) I only b) II only c) III only d) I and II only

3. Find the range of values of x if $2|x-3| > |3x+1|$

   a) $-\frac{9-10\sqrt{2}}{7} < x < \frac{9+10\sqrt{2}}{7}$ b) $-7 < x < 1$ c) $x < -7$ d) $-1 < x < 7$

4. Which of the following inequality is correct for all $a, b \in \mathbb{R}$

a) $|a-b| \leq |a| - |b|$

b) $|a-b| < |a| - |b|$

c) $|a-b| \geq |a| - |b|$

d) $|a-b| > |a| - |b|$

5. The complex number z is given by $z = \sqrt{3} + i$, express in cartesian form $\frac{i\bar{z}}{z}$ where $\bar{z}$ is the complex conjugate of z.

a) $\frac{1}{2} - i\frac{\sqrt{3}}{2}$

b) $\frac{1}{2} + i\frac{\sqrt{3}}{2}$

c) $\frac{\sqrt{3}}{2} - i\frac{1}{2}$

d) $\frac{\sqrt{3}}{2} + i\frac{1}{2}$

6. Simplify the expression $\frac{\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)^2}{\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)}$

a) $\frac{1}{2} + \frac{\sqrt{3}}{2}i$

b) $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$

c) $-\frac{\sqrt{3}}{2} + \frac{1}{2}i$

d) $-\frac{\sqrt{3}}{2} - \frac{1}{2}i$

7. Evaluate $\lim_{x\to 2} \frac{\sqrt{x^2+5}-3}{x^2-2x}$

a) $\frac{0}{0}$

b) $\frac{1}{3}$

c) 0

d) 3

8. Complete the following identities in hyperbolic functions

a) $\cosh^2 x - \sinh^2 x \equiv$ .... [0.5marks]

b) $\mathrm{cosech}^2 x \equiv$ .... [0.5marks]

c) $\cosh 2x \equiv$ .... [0.5marks]

d) $\tanh 2x \equiv$ .... [0.5marks]

9. Describe geometrically the set of points in the complex plane that satisfies

$2|z-3i| = |z|$

a) A circle with center (0,6) radius $3\sqrt{2}$ b) A circle with center (0,4), radius $3\sqrt{2}$

c) A circle with center (0,4) radius 2 d) A circle with center (0,6) radius 2

10. Does the $\lim_{x\to 1} f(x)$ exists for

$$f(x) = \begin{cases} 4 - x^2, & if\ x \leq 1 \\ 2 + x^2, & if\ x > 1 \end{cases} \ ?$$

If yes, what is the value?

a) No b) Yes and is equal to 2

c) Yes and is equal to 4 d) Yes and is equal to 3

11. If $f(x,y) = tan^{-1}(x+2y)$, find f_y.

a) $2sec^2(x+2y)$ b) $\frac{1}{1+(x+2y)^2}$ c) $\frac{2}{1+(x+2y)^2}$ d) $sec^2(x+2y)$

12. Find $\lim_{x\to 0} \frac{\tan 3x}{x\sqrt{3}}$

a) $\frac{0}{0}$ b) $\sqrt{3}$ c) $\frac{3}{\sqrt{3}}$ d) $\frac{1}{\sqrt{3}}$

13. Solve for x in $5coshx + 7sinhx = 1$

a) $x = \frac{1}{2}$ and $x = -\frac{1}{3}$ b) $x = In\left(\frac{1}{2}\right)$

c) $x = In\left(\frac{1}{2}\right)$ and $x = -In\left(\frac{1}{3}\right)$ d) $x = -In\left(\frac{1}{2}\right)$ and $x = In\left(\frac{1}{3}\right)$

14. What is the derivative of $In\left(tan\frac{1}{2}x\right)$?

a) $cosecx$ b) $\frac{1}{2}sec^2\frac{1}{2}x$ c) $\frac{sec^2\frac{1}{2}x}{tan\frac{1}{2}x}$ d) $\frac{tan\frac{1}{2}x}{sec^2\frac{1}{2}x}$

15. If $n \in \mathbb{R}$ and z is a complex number represented by $x + iy$ with principal value θ and modulus r. Complete the following:

a) Polar form of z : .... [1 mark]

b) Euler's form of z: ....[1 mark]

c) Polar form of z^n .... [1 mark]

d) Polar form $z^{1/n}$ .... [1 mark]

16. The function $f(x) = \frac{x}{x^3 - x}$ is an/a

a) Odd function b) Even function c) Periodic function d) None

17. Find the first derivative of $sinx = e^y$

a) $y' = cosx$ b) $y' = e^y cosx$ c) $e^y = cosx$ d) $y' = cotx$

18. Find the gradient of the curve $xsiny = \frac{1}{2}$ at $\left(1, \frac{\pi}{6}\right)$.

a) $-\frac{\sqrt{3}}{3}$ b) $-\frac{\sqrt{3}}{2}$ c) $\frac{2\sqrt{3}}{3}$ d) $\frac{-2\sqrt{3}}{3}$

19. The period of $f(x) = sin\left(\frac{\pi - \theta}{4}\right)$ is ?

a) -2π b) $\frac{2\pi}{4}$ c) -8π d) 8π

20. Find the numerical value of $\tanh(In\sqrt{3})$

a) 0 b) $\frac{1}{2}$ c) 1 d) 2

21. Write the total differential of the following.

a) Let $z = f(x, y)$ then,

$dz =$ ....[1mark]

b) Let $z = f(x, y)$ with $x = h(t)$ and $y = g(t)$ such that z is a composite function of t, then

$\frac{dz}{dt} =$ .... [1mark]

c) Let $z = f(x, y)$ with $x = h(u, v)$ and $y = g(u, v)$ such that z is a composite function of u and v then

$\frac{\partial z}{\partial u} =$ .... [1mark]

$\frac{\partial z}{\partial v} =$ .... [1mark]

22. If $z = u^2 + v^2$ and $u = at^2$, $v = 2at$, find $\frac{dz}{dt}$

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....[2 marks]

23. Write down the **names** of the following.

a) $\nabla =$ .... [0.5 marks]

b) $\nabla \phi =$ .... [0.5 marks]

c) $\nabla . \phi =$ .... [0.5 marks]

d) $\nabla \times \phi =$ .... [0.5 marks]

e) $\nabla^2\phi =$ ....[0.5 marks]

24. The differential operators in Que.(23) above have applications in the study of

a) ....

b) .... [0.5 marks]

.... [0.5 marks]

25. Evaluate $\lim_{n\to 0}\frac{\sqrt{n+3}-\sqrt{3}}{n}$

a) $\sqrt{3}$ b) $\frac{1}{2\sqrt{3}}$ c) 0 d) 3

26. If $f = x^2zi - 2y^3z^2j + xy^2k$. Find the curl of f at $(1,-1,1)$.

a) -3 b) $-6i$ c) $2xy - 6y^2z^2 + xy^3$ d) $(2xyz + 4y^3)i + (x^2 - y^2z)j$

27. Which of the following is not true about hyperbolic functions

a) $tanhx = \frac{e^{2x}-1}{e^{2x}+1}$ b) $sechx = \frac{2e^x}{e^{2x}+1}$ c) $sinhx = \frac{e^{-x}-e^x}{2}$ d) $coshx = \frac{e^{-x}+e^x}{2}$

28. The modulus of $\frac{2+3i}{1-i}$ is

a) $\frac{1}{2}\sqrt{26}$ b) $\frac{5}{2}+\frac{5}{2}i$ c) $-\frac{1}{2}+\frac{5}{2}i$ d) $\frac{1}{2}\sqrt{50}$

29. The complex factors of $4x^2 + 9$ is

a) $(2x+3)(2x-3)$ b) $(2x+3i)(2x-3i)$ c) $\left(x+\frac{3i}{2}\right)\left(x-\frac{3i}{2}\right)$ d) $(2x+3i)^2$

30. What is the derivative of $\sqrt{25-x^2}$?

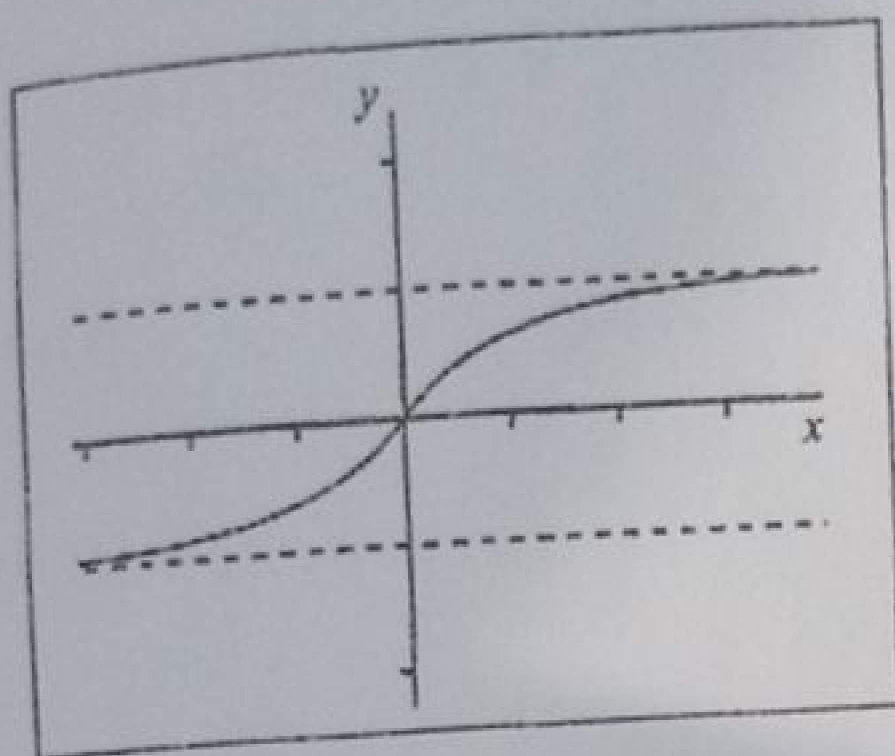
a) $\frac{1}{2\sqrt{25-x^2}}$ b) $\frac{1}{2}\sqrt{25-x^2}$ c) $-\frac{1}{2}x\sqrt{25-x^2}$ d) $\frac{-x}{\sqrt{25-x^2}}$

31. Find the value of $c \in [-1,\ 4]$ for which $f(x) = \frac{x^2}{4} + 1$ satisfies the hypothesis of the mean value theorem.

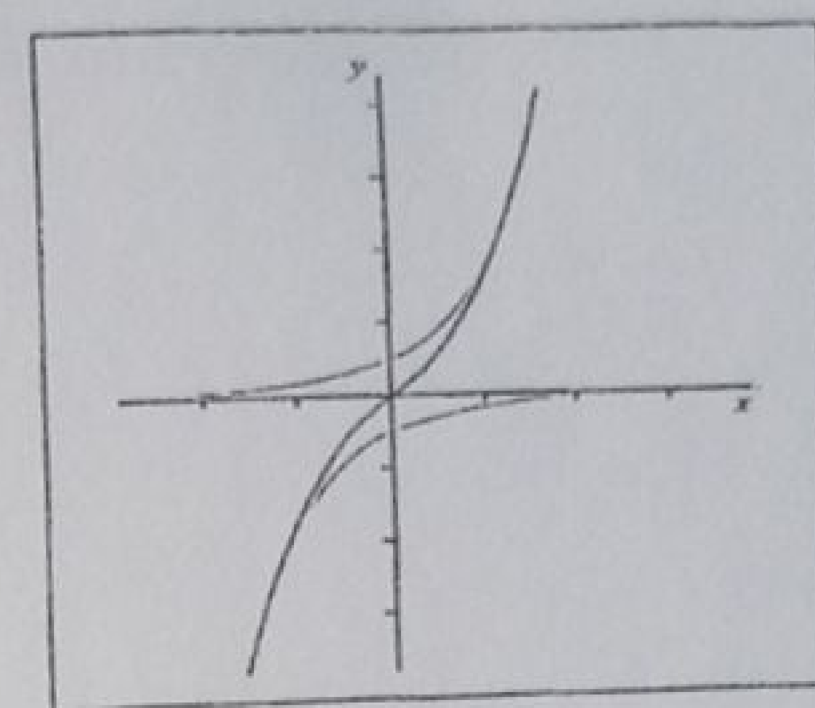
a) $\frac{3}{2}$ b) $\frac{3}{10}$ c) $\frac{5}{2}$ d) $\frac{3}{8}$

Which graph best describes $tanhx$?

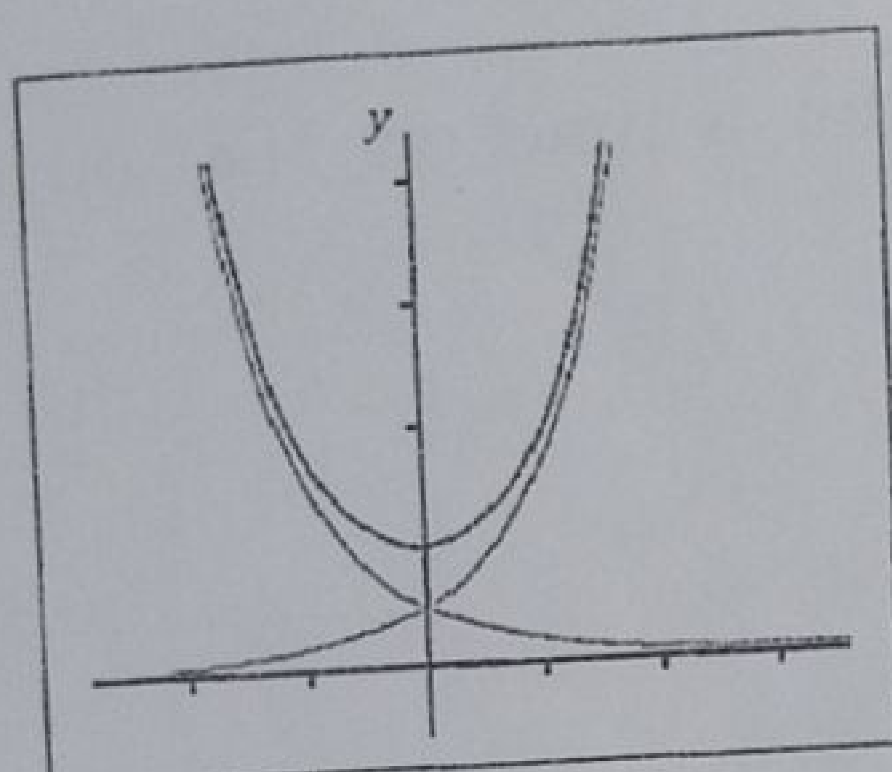
a)



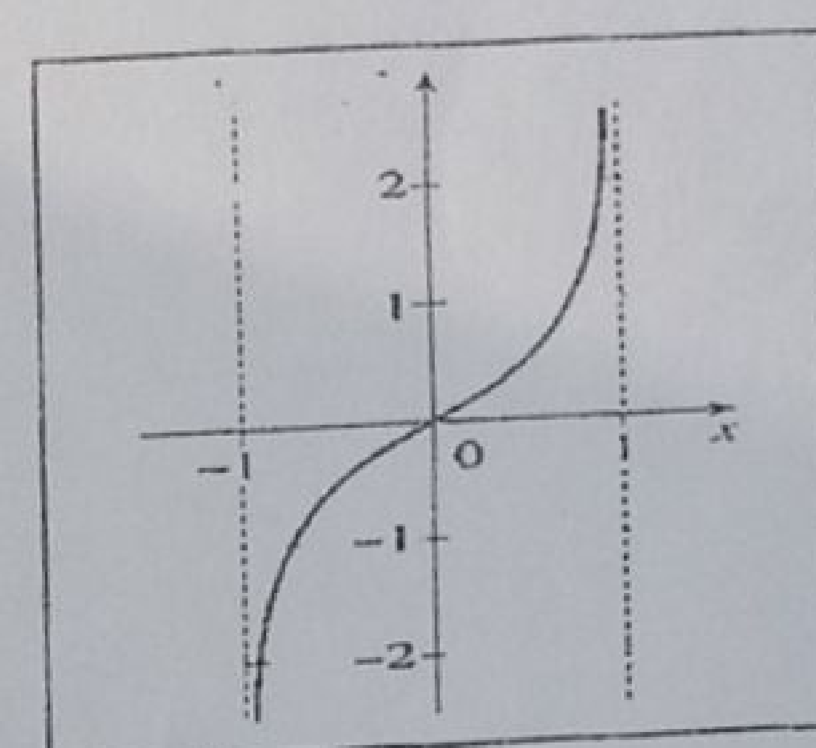
b)



c)



d)



33. Let $x, y \in \mathbb{R}$ such that $x < y < 0$. Also let $|x|$ denote the modulus of a function. Complete the diagram below by mapping appropriately.

Closed interval	(x, y)
Open interval	$(x, y]$
$\lvert x\rvert$	$x^2 > y^2$
$\lvert x+y\rvert$	$[x, y]$
Half closed interval	$\equiv \sqrt{x^2}$
$\lvert x\rvert > \lvert y\rvert$	$\leq \lvert x\rvert + \lvert y\rvert$

[0.5 mark each]

34. A function $f(x)$ is said to be continuous at the point $x = a$ if it satisfies the following hypothesis. **Write** these **hypothesis.**

a) ....[0.5mark]

b) ....[0.5mark]

c) ....[0.5mark]

35. **Write** the **hypothesis** of the Rolle's theorem for a function $f(x)$ on the closed interval $[a, b]$.

a) ....[0.5mark]

b) ....[0.5mark]

c) ....[0.5mark]

36. Let $y = uv$ where u and v are differentiable functions of x. Use Leibnitz's rule for successive differentiation to find the following derivatives. The first one has been done for you. Write your answer in it simplest form.

a) $\frac{dy}{dx} = \quad u\frac{dv}{dx} + v\frac{du}{dx}$

b) $\frac{d^2y}{dx^2} =$.. [1mark]

c) $\frac{d^3y}{dx^3} =$.. [1mark]

d) $\frac{d^ny}{dx^n} =$.. [1mark]

UNIVERSITY OF ENERGY AND NATURAL RESOURCES

SCHOOL OF SCIENCES

DEPARTMENT OF MATHEMATICS AND STATISTICS

End of First Semester Examinations , 2017/2018

B.Sc. (Pet./Ren. En./Elect./ Mech. /Env. Eng./ Agric./Comp./Civ. Eng.)

Candidates Programme:

Index Number:

ENGINEERING MATHS I

MATH 109

12Th DECEMBER, 2017

LEVEL 100

3.00 HOURS

Additional Materials: Non-programmable Calculator

INSTRUCTIONS

- ☐ Write your index number and programme of study in the spaces provided on the question booklet before starting the examination.
- ☐ There are 15 printed pages excluding back page and make sure you have all pages before starting the examination.
- ☐ The paper consists of two sections. Section A and Section B.
- ☐ Answer **all** questions in **Section A** on the question paper by **circling** the correct answer or **solving** the required problem in the spaces provided.
- ☐ Answer only **ONE** question from **Section B** in the answer booklet.
- ☐ **No** part of this question booklet should be taken out.

E.K. Donkoh

Section A (45% Marks)

1. On a sketch in an Argand diagram shade the region representing $|z - 2 + 2i| \leq 2$

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.... [2 marks]

2. Solve the equation $|4 - 2^x| = 10$, giving your answer correct to 3 $s.f.$

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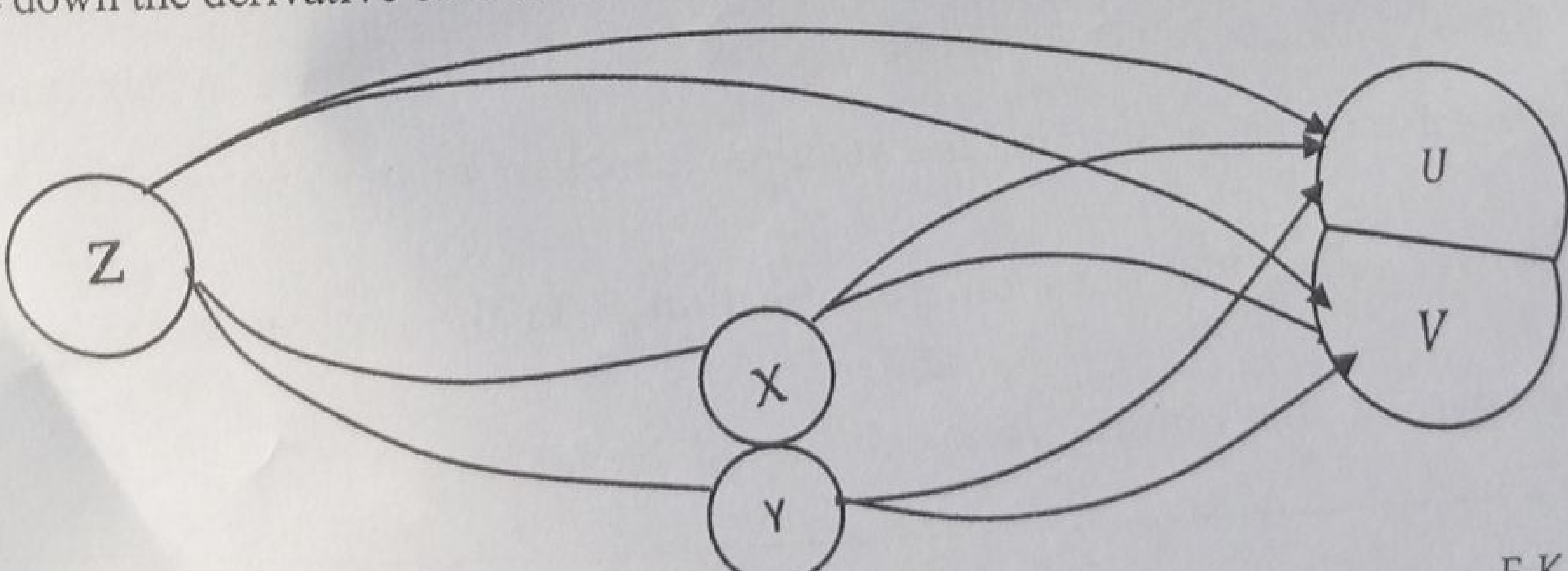
....[3 marks]

3. Write down the derivative of each route in the diagram below and find



E.K. Donkoh

a) $\frac{\partial z}{\partial u} =$ .... [1 mark]

b) $\frac{\partial z}{\partial v} =$ .... [1 mark]

4. The truth set of $6x^2 - x - 35 \geq 0$ is

a) $\left\{x: x \leq \frac{-7}{3} \cup x \geq \frac{5}{2}\right\}$ b) $\left\{x: \frac{7}{3} \leq x \leq \frac{-5}{2}\right\}$ c) $\left\{x: \frac{-7}{3} \leq x \leq \frac{-5}{2}\right\}$ d) $\left\{x: x \leq \frac{7}{3} \cup x \geq \frac{-5}{2}\right\}$

5. Evaluate i^{2016}

a) i b) $-i$ c) 1 d) -1

6. Find the value of $2\text{Re}z - 3\text{Im}z$ if $z = -4 - 5i$.

a) -7 $b)$ 23 $c)$ 7 $d)$ -4 $e)$ None of the above

7. If $z = cos\theta + isin\theta$, then $z^n - z^{-n}$ is

a) $2cos\, n\theta$ b) $\cos n\theta$ c) $2isin\, n\theta$ d) $isin\, n\theta$

8. Evaluate $\lim_{x \to 2} \frac{\sqrt{x^2+5}-3}{x^2-2x}$

a) $\frac{0}{0}$ b) $\frac{1}{3}$ c) 0 d) 3

9. Complete by simplifying the following

a) $|x_1 - x_2 - \cdots \ldots . -x_n| =$ .... [1 mark]

b) $|x_1 + x_2 + \cdots \ldots . +x_n| =$ .... [1 marks]

c) $|x_1 \times x_2 \times \ldots \ldots . \times x_n| =$ .... [1 mark]

10. If ϕ is an empty set, complete the sentence using the following

$\mathbb{R}$, $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{C}$

$\phi \subset$ .... $\subset$ .... $\subset$ .... $\subset$ ....$\subset$.............. [2.5marks]

11. Complete the following identities in the **form of exponential functions.**

a) $cosh^2 x - sinh^2 x \equiv$ ....[0.5mark]

b) $\sinh(x+y) \equiv$ .... [1marks]

c) $tanh2x \equiv$ ....

....[1mark]

12. Determine whether the $\lim_{x\to 4} f(x)$ exists for

$$f(x) = \begin{cases} 2x+1, & 0 \le x \le 2 \\ 7-x, & 2 < x < 4 \\ x\ , & 4 \le x \le 6 \end{cases} \ ?$$

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.... [4 marks]

13. The function $f(x) = x\frac{e^x+1}{e^x-1}$ is an/a

a) Odd function b) Even function c) Periodic function d) None

14. Express in polar form the complex number $1 - \sqrt{2} + i$

Answer: ....

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.... [3 mark]

15. One of the maxima and minima values of $f(x,y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$ is

a) $(\sqrt{2}, -\sqrt{2})$ b) $(2,-2)$ c) $(\sqrt{2}, \sqrt{2})$ d) $(-2,2)$

16. Find the gradient of the curve $x^3 - 2y^3 = 3xy$ at $(2,1)$.

b) $\frac{3}{4}$ b) $-\frac{1}{2}$ c) $\frac{1}{2}$ d) $\frac{5}{4}$

17. Which of the following best describes the Clairauts theorem

a) Suppose that f is defined on a disk D that contains the point (a,b).If the functions f_{xy} and f_{yx} are continuous on the disk then $f_{xx} = f_{yx}$.

b) Suppose that f is defined on a disk D that contains the point (a,b).If the functions f_{xy} and f_{yx} are continuous on the disk then $f_{xy} = f_{yx}$.

c) Suppose that f is defined on a disk D that contains the point (a,b).If the functions f_{xy} and f_{yx} are continuous on the disk then $f_{xx} = f_{yy}$.

d) Suppose that f is defined on a disk D that contains the point (a, b).If the functions f_{xy} and f_{yx} are continuous on the disk then $f_{xy} = f_{yy}$.

18. Find the numerical value of $\cosh(In\sqrt{3})$

a) 0 b) $\sqrt{3}$ c) $\frac{2}{\sqrt{3}}$ d) $\frac{1}{\sqrt{3}}$

19. Evaluate the following

a) $\lim_{h \to 0} \frac{2-\sqrt{4+h}}{2}$

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....[2 marks]

b) $\lim_{h \to -\infty} \left(\frac{3x}{x-1} - \frac{2x}{x+1}\right)$

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....[2 marks]

20. Find $\left(\frac{1}{2} + \frac{1}{2}i\right)^{10}$

a) $\frac{1}{16}i$ b) $\frac{1}{32}i$ c) $\frac{1}{64}i$ d) $\frac{1}{18}i$

21. The Euler form of the complex number $-5 + 5i$ is

a) $5\sqrt{2}e^{i\pi/4}$ b) $5\sqrt{2}e^{-i3\pi/4}$ c) $5\sqrt{2}e^{i3\pi/4}$ d) None of these

22. If $z = x^2y^3$ and $x = 2t^3$, $y = 3t^2$, find $\frac{dz}{dt}$

....

....[2 marks]

23. Write down the notation for the following.

a) Laplacian of ϕ .... [0.5 marks]

b) Hamiltonian of ϕ in 3 - D.. [0.5 marks]

24. Write down the total differential of

$$z = f(x_1, x_2, \dots, x_n)$$

....

....[1 mark]

25. The function $f(x) = \begin{cases} \frac{x^2-4}{x-2}, & if\ x \neq 2 \\ C, & if\ x \neq 2 \end{cases}$ is continuous. What is the value of C?

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[3 marks]

26. Evaluate the following

a) $\lim_{x \to 3^-} \left(\sqrt{9 - x^2} + x\right) =$

[1 mark]

b) $\lim_{x \to +\infty} \frac{7x-4}{\sqrt{x^3+5}} =$

[1 mark]

27. The modulus of $\frac{2+3i}{1-i}$ is

a) $\frac{1}{2}\sqrt{26}$ b) $\frac{5}{2} + \frac{5}{2}i$ c) $-\frac{1}{2} + \frac{5}{2}i$ d) $\frac{1}{2}\sqrt{50}$

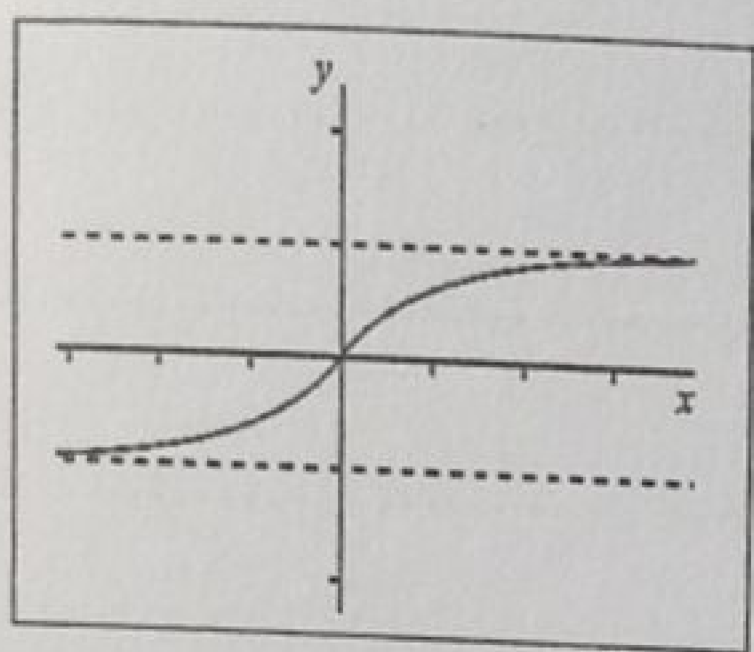
28. Write down the complex factors for $4x^2 - 12x + 25$

....

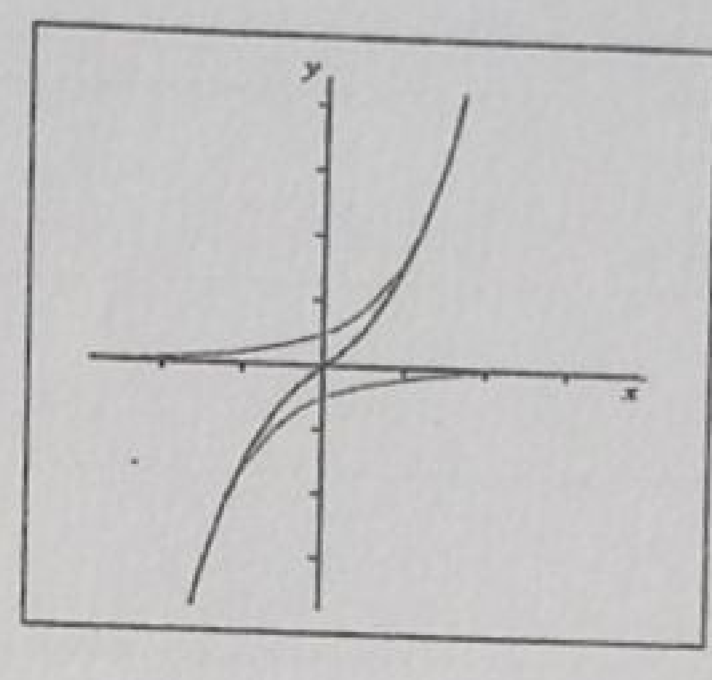
....[2marks].

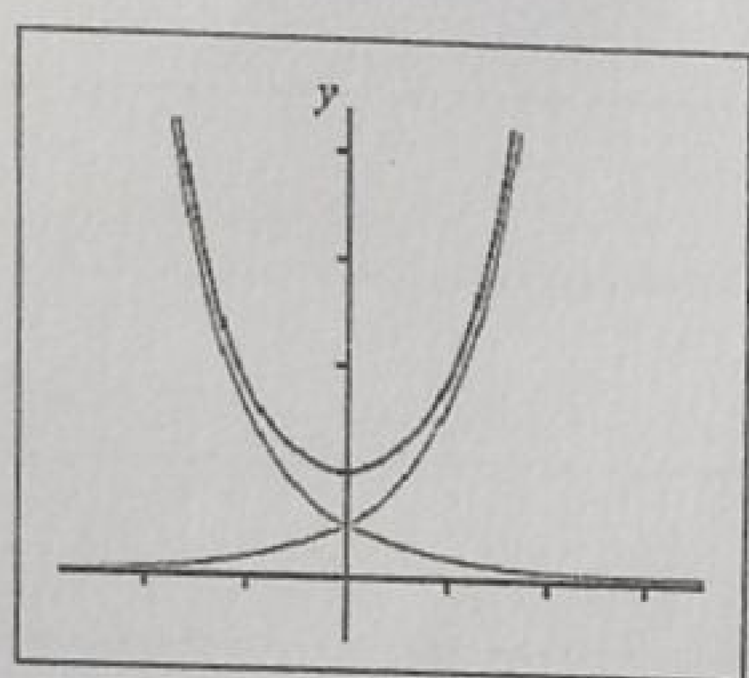
29. Which graph best describes *sinhx* ?

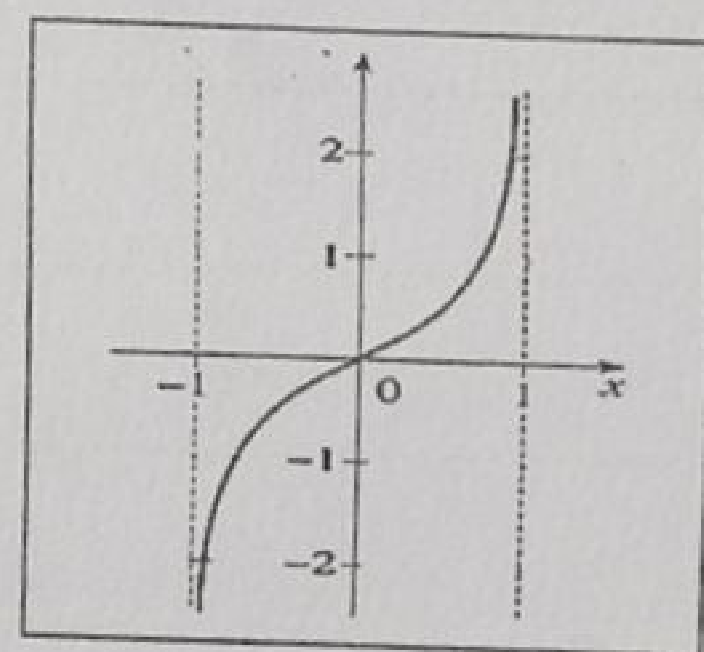
a)



b)







30. Use L' Hopitals rule to find

$$\lim_{x \to +\infty} \frac{In(1+x)}{1+x}$$

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.... [2 mark]

31. Find the number c such that $f(x) = Ax^2 + bx + c$ satisfy the mean value theorem on the closed interval $[a, b]$.

[3 marks]

32. Any point of the curve $x = t^2 - 1,\ y = 4t - 3,\ z = 2t^2 - 6t$. Find

a) The space curve [1 mark]

b) The tangent vector

[1 mark]

31. Find the number c such that $f(x) = Ax^2 + bx + c$ satisfy the **mean value theorem** on the closed interval $[a, b]$.

[3 marks]

32. Any point of the curve $x = t^2 - 1,\ y = 4t - 3,\ z = 2t^2 - 6t$. Find

a) The space curve [1 mark]

b) The tangent vector

[1 mark]

c) Unit tangent vector

[2 marks]

d) The normal vector

[2marks]

33. If $A = x^2zi - 2y^3z^2j + xy^2k$. Find the curl of A at $(1, -1, 1)$.

a) -3 b) $-6i$ c) $2xy - 6y^2z^2 + xy^3$ d) $(2xyz + 4y^3)i + (x^2 - y^2z)j$

d) $tan(e^{y-x}) = x^2$

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....[3mark]

37. Find the fundamental period of $f(t) = \sin(10t - 1) + 2\cos(5t - 2)$

Answer:...

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.... [3 marks]

SECTION B

Answer **Only One** Question in Section B (15% Marks)

Provide your Answers in the Answer Booklet.

43. a)Prove by induction that $5^n + 12n - 1$ is divisible by 16. [5 marks]

b) Find the roots of the equation $z^3 + 4\sqrt{3} = 4i$, giving your answers in the form $re^{-i\theta}$ where $r > 0$ and $0 < \theta < 2\pi$ [7 marks]

c) Prove that $\sqrt{2}$ is irrational. [3 marks]

44. a) If $f(x) = x^{\sin x}$ for $x > 0$,find $f'(x)$ [3 marks]

b) Verify Clairuat's theorem for $f(x, y) = ye^{-x^2y^2}$ [5 marks]

c) Compute the total differential for $f(x, y) = e^{x^2+y^2} \sin 2x$ [3 marks]

d) Use Leibnitz's rule to evaluate $\frac{d^n}{dx^n}(x^4e^{2x})$. [4 marks]

45. a) The complex numbers u and w satisfy the equations

$$u - w = 4i \quad \text{and} \quad uw = 5$$

Solve the equations for u and w, giving all answers in the form $x + iy$, where x and y are real. [5 marks]

a) Given that $'a'$ is a constant, prove by mathematical induction that, for every positive integer n, $\frac{d^n}{dx^n}(xe^{ax}) = na^{n-1}e^{ax} + a^nxe^{ax}$. [6 marks]

b) Prove that $(coshx - sinhx)^n \equiv coshnx - sinhnx$ [4 marks]

E.K. Donkoh

UNIVERSITY OF ENERGY AND NATURAL RESOURCES

SCHOOL OF SCIENCES

DEPARTMENT OF MATHEMATICS AND STATISTICS

End of First Semester Examinations , 2019/2020

B.Sc. (Agric./Mech./ Env./Ren./ Civil/Petr./Comp./Elect. Engineering)

Candidates Programme:

Index Number:

ENGINEERING MATHS I — MATH 109

11^{TH} DECEMBER, 2019

LEVEL 100 — 2.30 HOURS

Additional Materials: Non-programmable Calculator

INSTRUCTIONS

☐ Write your index number and programme of study in the spaces provided on the question booklet before starting the examination.

☐ There are 8 printed pages excluding back page and make sure you have all pages before starting the examination.

☐ The paper consists of three sections. Section A , B and Section C.

☐ Answer **all** questions in **Section A** and B on the question paper by **circling** the correct answer or **solving** the required problem in the spaces provided.

☐ Answer only **ONE** question from **Section C** in the answer booklet.

☐ **No** part of this question booklet should be taken out.

SECTION A (20% Marks)

1. Fill in the following
   a) $\infty - \infty = \cdots$ .... [0.5 mark]
   b) $e^{\infty} =$ .... [0.5 mark]
   c) $\frac{\infty}{\infty} =$ .... [0.5 mark]
   d) $\frac{0}{0} =$ .... [0.5 mark]
   e) $\frac{1}{\infty} =$ .... [0.5 mark]

2. Evaluate i^{2016}

   a) i b) $-i$ c) 1 d) -1

3. The inequality $(2z+1)^2 > 9$ is satisfy when
   a) $z > 1$ or $z < -2$
   b) $z < 1$ or $z < -2$
   c) $z < 1$ or $z > -2$
   d) $z > 1$ or $z > -2$

4. Using *De' Moivre's theorem* if $z = cos\theta + isin\theta$, then $z^n - z^{-n}$ is

   a) $2cos\, n\theta$ b) $\cos n\theta$ c) $2isin\, n\theta$ d) $isin\, n\theta$

5. *Complete* by simplifying the following
   a) $\left|\frac{x}{y}\right| =$ .... [0.5 marks]
   b) $|x+y|$ .... [1 marks]
   c) $|x_1 \times x_2 \times \ldots\ldots \times x_n| =$ .... [1 mark]
   d) $|x|^2 \equiv$ .... [0.5mark]

6. If ϕ is an empty set, complete the sentence using the following

   $\mathbb{R}$, $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{C}$

   $\phi \subset$ .... $\subset$ .... $\subset$ .... $\subset$ ....$\subset$......... [2.5marks]

7. The period of $cos\left(\frac{1-2x}{5}\right)$ is ?

   a) 2π b) $\frac{2\pi}{5}$ c) -5π d) 5π

8. Which of the following defines for f for which $f(-x) = f(x)$.
   a) $y = xsinx$
   b) $y = sinx$
   c) $y = xcosx$
   d) $y = Inx$

9. Let $z_1 = r_1e^{i\alpha}$ and $z_2 = r_2e^{i\beta}$. All these is/are true about a complex number z_1, z_2 except

   i. $z_1 + z_2 = r_1r_2e^{i(\alpha+\beta)}$

   ii. $\frac{z_1}{z_2} = \frac{r_1}{r_2}e^{i(\alpha-\beta)}$

   iii. $z_1z_2 = r_1r_2e^{i(\alpha-\beta)}$

   A. I only

   B. I, II and III

   C. I and III only

   D. III only

10. Let $x, y \in \mathbb{R}$ such that $x < y$. Also let $|x|$ denote the modulus of a function. Complete the diagram below by mapping appropriately. [0.5marks each]

Closed interval	(x, y)
Open interval	$(x, y]$
$\lvert x\rvert$	$x^2 > y^2$
$\lvert x + y\rvert$	$[x, y]$
Half closed interval	$\equiv \sqrt{x^2}$
$\lvert x\rvert > \lvert y\rvert$	$\leq \lvert x\rvert + \lvert y\rvert$

11. Given that $2 + 3i = \sqrt{a + bi}$, the values of a and b is

a) $a = 2$ and $b = -3$

b) $a = 4$ and $b = 9$

c) $a = -5$ and $b = -12$

d) $a = -5$ and $b = -12$

12. Given that $z_1 = 3 + 2i$ and $z_2 = -1 - 4i$, find $Im(z_1 z_2)$..

a) $5 - 14i$

b) -14

c) i

d) 14

tion.

ach]

SECTION B (30% MARKS)

16. Write the derivative of the following?

a) $y = In(\sqrt{4x+5})$

[1.5 marks]

b) $cos^{-1}(1-2x^2)$

[3 marks]

c) $2x^2y + 3x^3 = siny$

[3 marks]

d) $\frac{3y}{2x}$

[2 marks]

17. Find the numerical value of $\sinh(In2)$

[2 marks]

18. Solve the equation $\frac{x^2-1}{x^2+1} = tanh(In2)$

[3 marks]

19. A function $f(x)$ is defined as $f(x) = \frac{x^3+1}{X+1}, x \neq -1$. Find $\lim_{x \to -1} f(x)$

[3 marks]

20. E

20. Evaluate the following

a) $\lim_{x \to 3^-} \left(\sqrt{9 - x^2} + x\right)$

b) $\lim_{x \to +\infty} \frac{7x-4}{\sqrt{x^3+5}}$

[2 marks]

[2marks]

21. Use L′ Hopital′s rule to evaluate the following

a) $\lim_{x \to 0} \left(\frac{e^{2x} - 1}{sinx}\right)$

b) $\lim_{x \to +\infty} \left(\frac{x^2}{e^x}\right)$

17. Find the numerical value of $\sinh(In2)$

[2 marks]

18. Solve the equation $\frac{x^2-1}{x^2+1} = tanh(In2)$

[3 marks]

19. A function $f(x)$ is defined as $f(x) = \frac{x^3+1}{X+1}, x \neq -1$. Find $\lim_{x \to -1} f(x)$

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20. Evaluate the following

a) $\lim_{x\to 3^-}\left(\sqrt{9-x^2}+x\right)$

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[2 marks]

[2marks]

21. Use L' Hopital's rule to evaluate the following

a) $\lim_{x\to 0}\left(\frac{e^{2x}-1}{sinx}\right)$

b) $\lim_{x\to +\infty}\left(\frac{x^2}{e^x}\right)$

22. Use the hypothesis of the mean value theorem to find c in $f(x) = \frac{x+3}{x-4}$ on $[1,3]$.

[3marks]

23. Find the *modulus* and *argument* of $-1 - i\sqrt{3}$

Modulus

Argument

24. If $z = In(x^2 + y^2)$ and $x = e^{-t}$, $y = e^{t}$, find $\frac{dz}{dt}$.

[3 marks]

25. The curve C: $3^{x-1} + xy - y^2 + 5 = 0$. Find $\frac{dy}{dx}$ at points (1,3).

[4marks]

SECTION C (10% Mark)

Answer **Only One** Question (**Answer Booklet**)

1. a) Prove using Leibnitz rule of differentiation that

$$\frac{d^n}{dx^n}(xe^x) = (x+n)e^x.$$

[5 marks]

b) Find the *modulus and argument* of $(5+6i^{11})+(8i^3+i^5)+(i^2-i^4)$. [5marks]

c) Determine whether the *hypothesis of the Rolle's theorem* holds for

$$f(x) = In(x^2+3) - In(4x) \text{ on } [1,3].$$

[5 marks]

2. a) Find the relative maxima and minima of $f(x,y) = x^3+y^3-3x-12y+20$

[6marks]

b) If $A = 3xyz^2 i + 2xy^3 j - x^2yzk$ and $\phi = 3x^2 - yz$. Find at the point $(1,-1,3)$ the following

i. $\nabla \times A$ [2.5 marks] ii. $A.\nabla\phi$ [2.5 marks]

c) Find the unit vector normal to the surface $x^2y - 2xz + 2y^2z^4 = 10$ at the point $(2,1-1)$

[4 marks]

4. (a) Show that if $y = \tan^{-1}\left(\frac{\sin t}{\cos t - 1}\right)$ then $\frac{dy}{dt} = \frac{1}{2}$. (3 Marks)

(b) Prove by mathematical induction that $\frac{d^n}{dx^n}[\sin ax] = a^n \sin(ax + \frac{n\pi}{2})$ for $n = 1, 2, 3, , \cdots$. (6 Marks)

(c) Express in the form of $r(\cos\theta + i\sin\theta)$ the roots of the equation $z^3 - 16 = 0$ (3 Marks)

(d) Show that $\lim_{x\to 0} x^2 \cos\frac{1}{x} = 0$ (3 Marks)